4x4 signed analog matrix — DC to 1 MHz

Rev A2: complete output rows on single sheets; AD5544 footprint corrected to SSOP-28.
+/-1 V peak inputs; up to +/-4 V peak into an external 50 ohm load.
Global labels join the sheets. Power names +5A/-5A mean +/-5 V analog.
SPICE results and model limitations are in the accompanying report; hardware validation remains.

Inputs

File: inputs.kicad_sch

Row 0

File: row_0.kicad_sch

Row 1

File: row_1.kicad_sch

Row 2

File: row_2.kicad_sch

Row 3

File: row_3.kicad_sch

Offset Trim

File: offset_trim.kicad_sch

Control

File: control.kicad_sch

Power

File: power.kicad_sch

Sheet: /
File: analog_matrix_4x4.kicad_sch

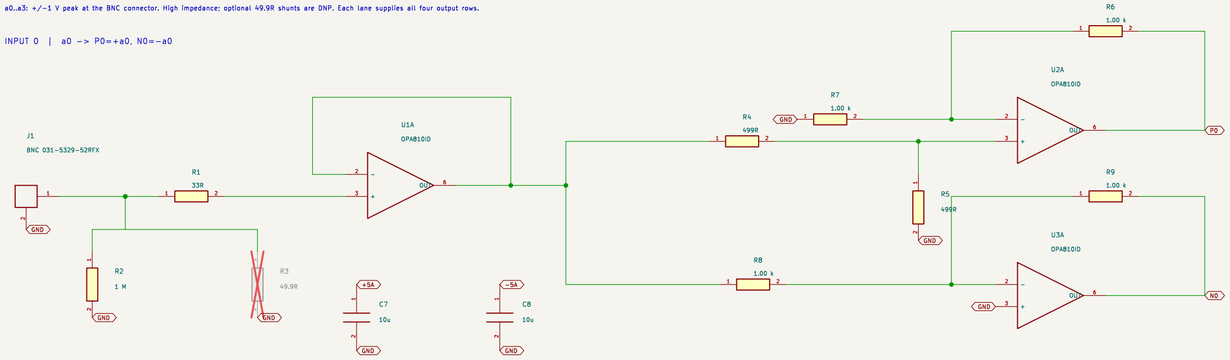
Title: 4x4 programmable analog matrix

Size: A3	Date: 2026-09-30	**Rev: A2**
KiCad E.D.A. 8.0.6		Id: 1/9

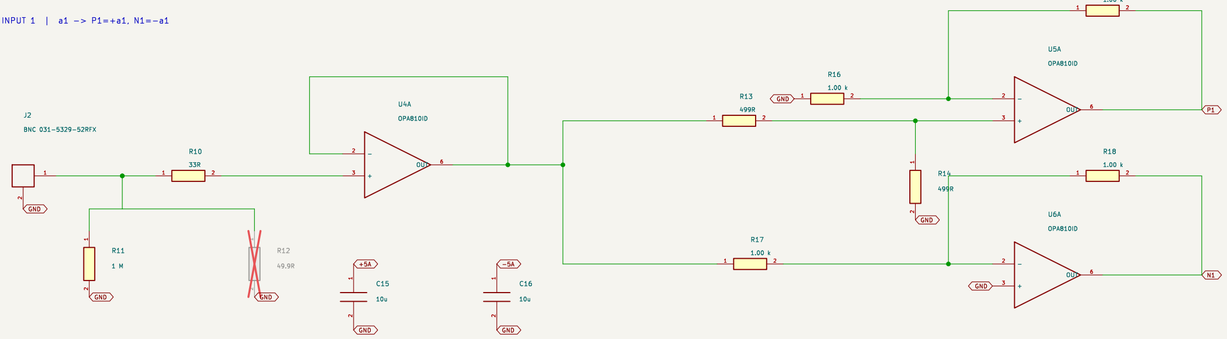
Four input buffers and matched positive / negative paths

a0..a3: +/−1 V peak at the BNC connector. High impedance; optional 49.9R shunts are DNP. Each lane supplies all four output rows.

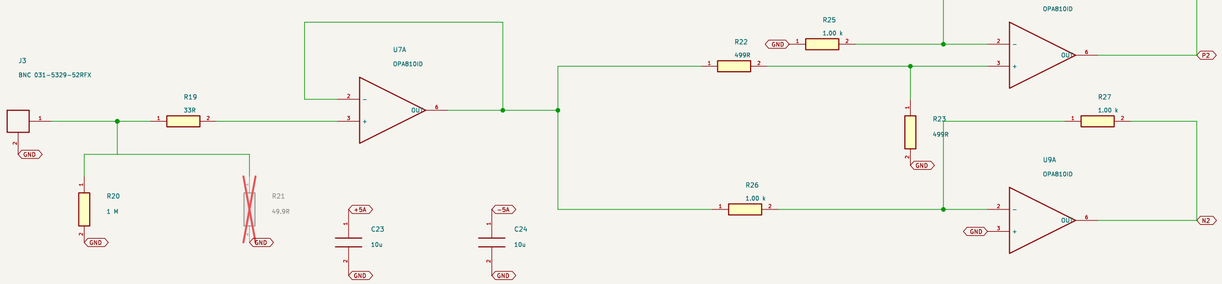
INPUT 0 | a0 -> P0=+a0, N0=−a0



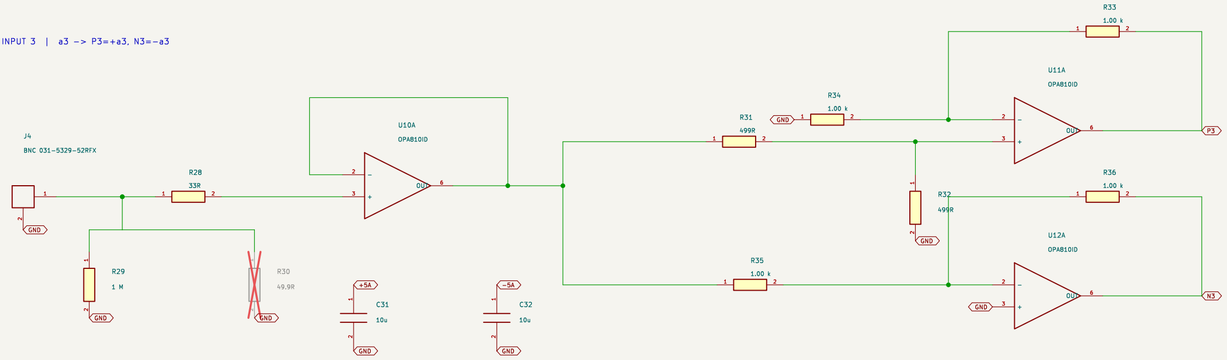
INPUT 1 | a1 -> P1=+a1, N1=−a1

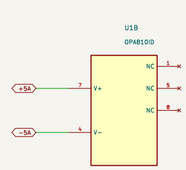


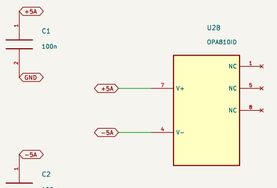
INPUT 2 | a2 -> P2=+a2, N2=−a2

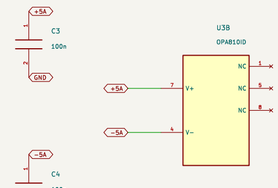


INPUT 3 | a3 -> P3=+a3, N3=−a3

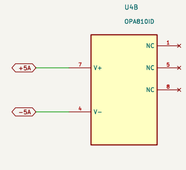


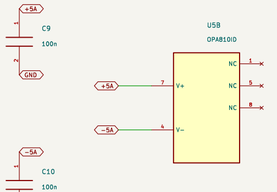


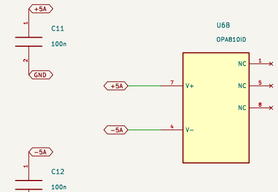




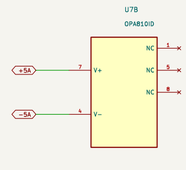


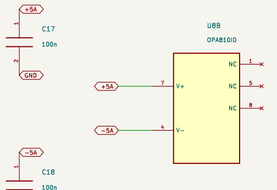


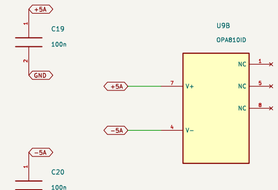




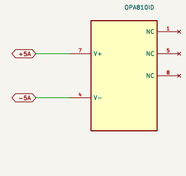


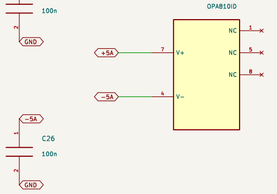


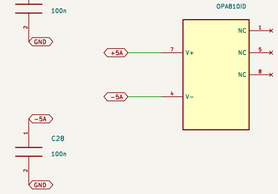














DC–1MHz; +/−1V input; +/−4V into 50 ohm. Prototype; not hardware validated.

Analog Matrix

Sheet: /Inputs/
File: inputs.kicad_sch

Title: Four input buffers and matched positive / negative paths

Size: A2	Date: 2026-09-30	**Rev: A2**
KiCad E.D.A. 8.0.6		Id: 2/9

Output b0: signed weights and 50-ohm driver

Follow left to right: polarity -> DAC magnitude -> current-to-voltage -> sum -> line driver. Each weight includes its own local power units below.

M[0,0] | input a0

M[0,1] | input a1

M[0,2] | input a2

M[0,3] | input a3

Branch isolation: 150R for loaded-buffer stability.
TIA: 2.2pF C0G; keep IOUT and FB copper short.
AD5544BRSZ: SSOP-28, 5.3 x 10.2 mm body.

SUM AND OUTPUT

49.9R >=1W. External load: 50R.
+/-4V peak at the loaded connector.
No C across THS3091 feedback.

DC-1MHz; +/-1V input; +/-4V into 50 ohm. Prototype; not hardware validated.

Analog Matrix	
Sheet: /Row 0/	
File: row_0.kicad_sch	
Title: Output b0: signed weights and 50-ohm driver	
Size: A2	Date: 2026-09-30
KiCad E.D.A. 8.0.6	**Rev: A2**
	Id: 3/9

Output b1: signed weights and 50-ohm driver

Follow left to right: polarity -> DAC magnitude -> current-to-voltage -> sum -> line driver. Each weight includes its own local power units below.

M[1,0] | input a0

M[1,1] | input a1

M[1,2] | input a2

M[1,3] | input a3

Branch isolation: 150R for loaded-buffer stability.
TIA: 2.2pF C0G; keep IOUT and FB copper short.
AD5544BRSZ: SSOP-28, 5.3 x 10.2 mm body.

SUM AND OUTPUT

49.9R >=1W. External load: 50R.
+/-4V peak at the loaded connector.
No C across THS3091 feedback.

DC-1MHz; +/-1V input; +/-4V into 50 ohm. Prototype; not hardware validated.

Analog Matrix	
Sheet: /Row 1/	
File: row_1.kicad_sch	
Title: Output b1: signed weights and 50-ohm driver	
Size: A2	Date: 2026-09-30
KiCad E.D.A. 8.0.6	**Rev: A2**
	Id: 4/9

Output b2: signed weights and 50-ohm driver

Follow left to right: polarity -> DAC magnitude -> current-to-voltage -> sum -> line driver. Each weight includes its own local power units below.

M[2,0] | input a0

M[2,1] | input a1

M[2,2] | input a2

M[2,3] | input a3

Branch isolation: 150R for loaded-buffer stability.
TIA: 2.2pF C0G; keep IOUT and FB copper short.
AD5544BRSZ: SSOP-28, 5.3 x 10.2 mm body.

SUM AND OUTPUT

49.9R >=1W. External load: 50R.
+/-4V peak at the loaded connector.
No C across THS3091 feedback.

DC-1MHz; +/-1V input; +/-4V into 50 ohm. Prototype; not hardware validated.

Analog Matrix	
Sheet: /Row 2/	
File: row_2.kicad_sch	
Title: Output b2: signed weights and 50-ohm driver	
Size: A2	Date: 2026-09-30
KiCad E.D.A. 8.0.6	**Rev: A2**
	Id: 5/9

Output b3: signed weights and 50-ohm driver

Follow left to right: polarity -> DAC magnitude -> current-to-voltage -> sum -> line driver. Each weight includes its own local power units below.

M[3,0] | input a0

M[3,1] | input a1

M[3,2] | input a2

M[3,3] | input a3

Branch isolation: 150R for loaded-buffer stability.
TIA: 2.2pF C0G; keep IOUT and FB copper short.
AD5544BRSZ: SSOP-28, 5.3 x 10.2 mm body.

SUM AND OUTPUT

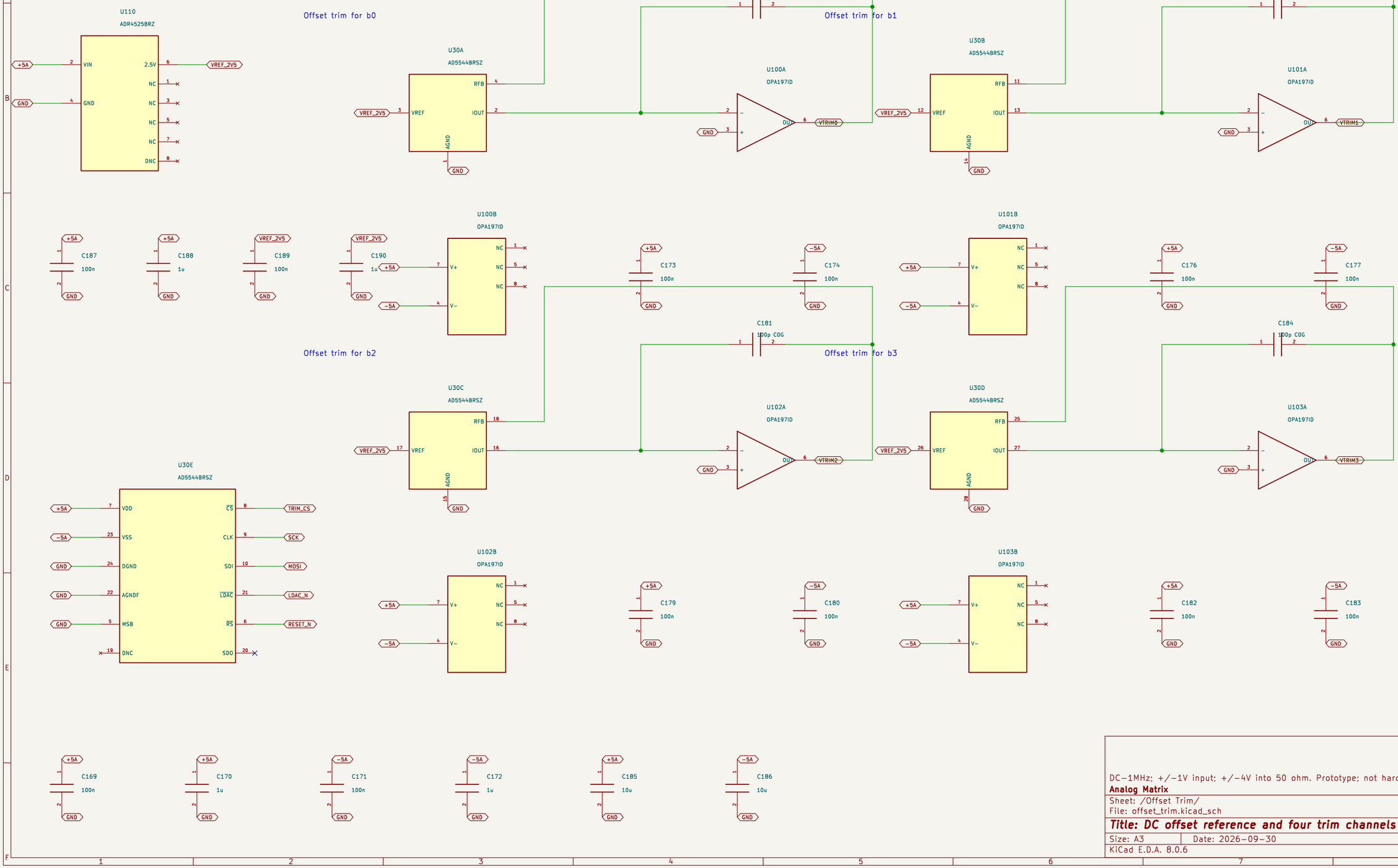
49.9R >=1W. External load: 50R.
+/-4V peak at the loaded connector.
No C across THS3091 feedback.

DC-1MHz; +/-1V input; +/-4V into 50 ohm. Prototype; not hardware validated.

Analog Matrix	
Sheet: /Row 3/	
File: row_3.kicad_sch	
Title: Output b3: signed weights and 50-ohm driver	
Size: A2	Date: 2026-09-30
KiCad E.D.A. 8.0.6	**Rev: A2**
	Id: 6/9

DC offset reference and four trim channels

U30 is a fifth AD5544. VTRIM=−2.5*D/65536 V. Midscale=32768. Loaded output trim is approximately +/−131 mV.



DC−1MHz; +/−1V input; +/−4V into 50 ohm. Prototype; not hardware validated.

Analog Matrix

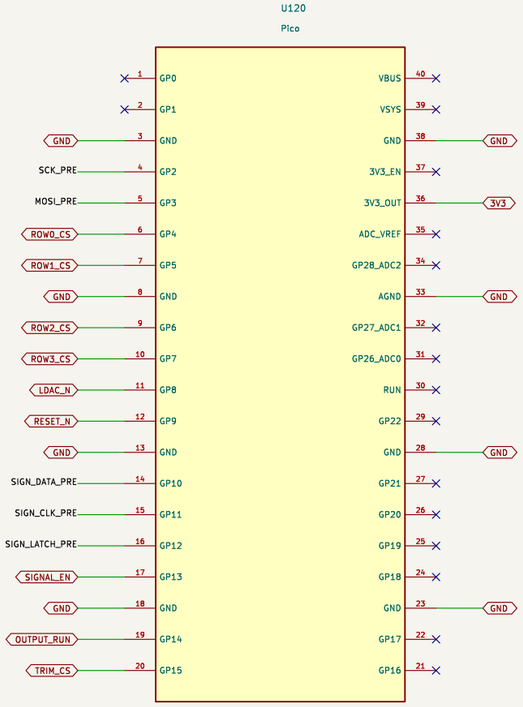
Sheet: /Offset Trim/
File: offset_trim.kicad_sch

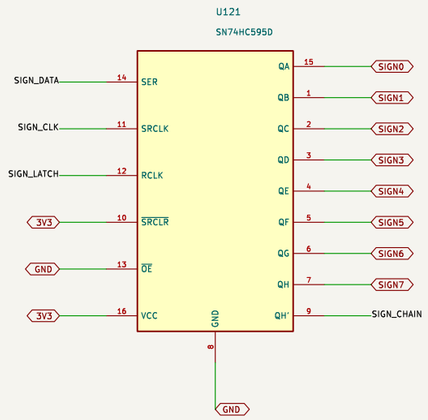
Title: DC offset reference and four trim channels

Size: A3	Date: 2026−09−30	**Rev: A2**
KiCad E.D.A. 8.0.6		Id: 7/9

RP2040 USB / SPI controller and sign-bit latches

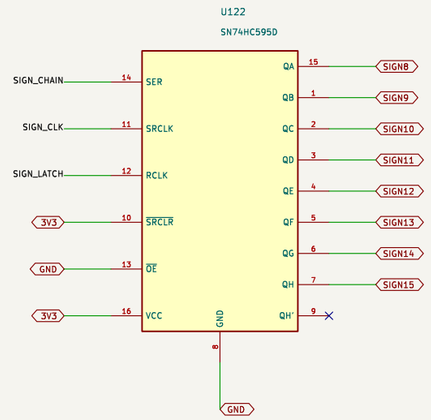
Pico is USB-powered. Do not connect VBUS/VSYS to +5A. Apply analog power first. GP14=OUTPUT_RUN and GP13=SIGNAL_EN default low.





First register QA..QH = SIGN0..7.
Second register QA..QH = SIGN8..15.
Clock bit 15 first, latch after 16 clocks.





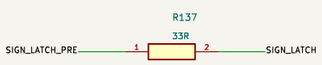
















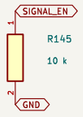


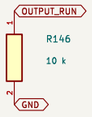












SPI mode 0, 1MHz, 24 clocks: last 18 bits = channel[1:0], code[15:0]. Five separate CS nets; shared LDAC. SDO pins intentionally unused.
Firmware sequence: mute -> disable sign switches -> latch signs -> load 20 DAC words -> LDAC -> settle -> restore prior armed state.
No automatic rail supervisor is fitted; static operation does not require continuous MCU traffic.

DC–1MHz; +/–1V input; +/–4V into 50 ohm. Prototype; not hardware validated.

Analog Matrix

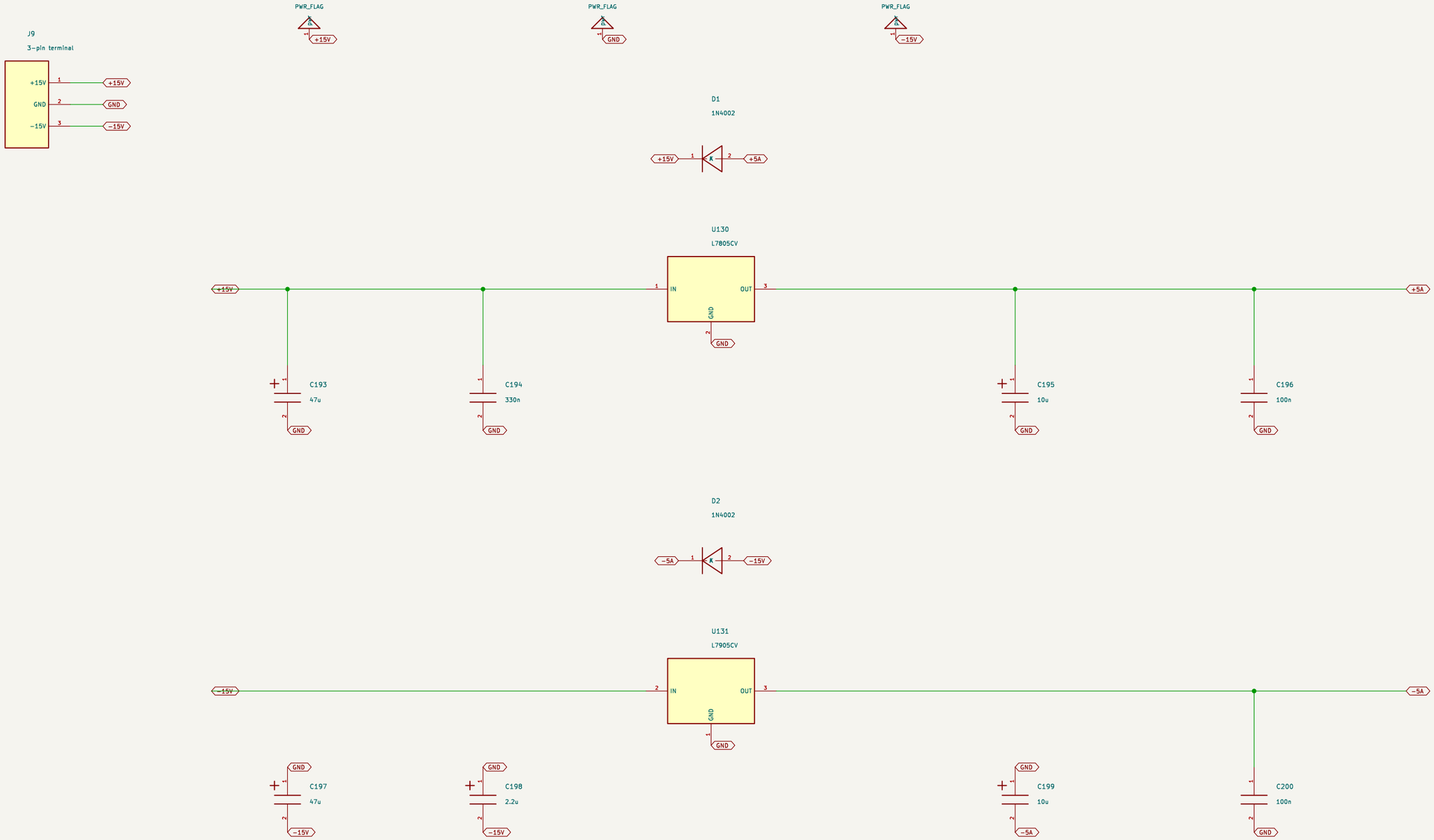
Sheet: /Control/
File: control.kicad_sch

Title: RP2040 USB / SPI controller and sign-bit latches

Size: A3	Date: 2026-09-30	**Rev: A2**
KiCad E.D.A. 8.0.6		Id: 8/9

External +/−15 V input and +/−5 V analog regulators

External regulated +/−15 V, at least 1 A per rail. Regulator heatsinks <=15 C/W. Negative regulator tab is −15 V; positive regulator tab is GND.



Observe all polarized capacitor signs. Use the shown electrolytic/tantalum regulator capacitors; do not substitute ceramic-only outputs. Regulator dissipation is roughly 3 W per device; verify temperature under sustained DC output loads. +/−5A supplies the analog core.

DC–1MHz; +/−1V input; +/−4V into 50 ohm. Prototype; not hardware validated.

Analog Matrix

Sheet: /Power/
File: power.kicad_sch

Title: External +/−15 V input and +/−5 V analog regulators

Size: A3	Date: 2026-09-30	**Rev: A2**
KiCad E.D.A. 8.0.6		Id: 9/9